MATHEMATICS
ANNOTATED READING NOTES

PDE

Partial Differential Equations

英文精读札记与中文边注

Prefaces and Chapter Outline

Reading companion to Lawrence C. Evans
Partial Differential Equations, Second Edition

基于 Evans 原著的学习整理

$$\partial_t u - \Delta u = 0$$

LUMENBOOK

2026-11-09

澄明讲义 / 数学阅读札记

How to Use These Notes

These notes follow Lawrence C. Evans's *Partial Differential Equations*, second edition. The English PDF supplies the chapter order and mathematical source. The supplied Chinese translation is a reference for terminology and reading support.

English text and margin notes

Read the main argument first. Margin notes will explain selected vocabulary, long sentences, and mathematical ideas. They will be short enough to remain beside the relevant passage. Longer derivations belong in the main text.

Expanded proofs

An expanded proof will state the assumptions, explain the strategy, and fill in omitted calculations. Each use of approximation, integration by parts, compactness, or passage to a limit must identify the applicable hypotheses. Source theorem numbers will be recorded explicitly; the notes' own numbering does not automatically coincide with the original book.

Current status

The two source prefaces have been typeset in English with selected Chinese margin notes. Chapters 1–12 and Appendices A–E remain outlines: their exposition, proofs, and exercises have not yet been transcribed or expanded. The margin notes on this page demonstrate the annotation system.

The introduction precedes Part I, as in the source. The three parts contain Chapters 2–4, 5–7, and 8–12. Appendices A–E supply background material. The bilingual glossary is an added study aid. Detailed subsection headings will be added when their source pages are studied.

中文提示

正文以英文为主，必要时用中文边注解释词句和数学难点。中译本辅助理解；遇到版本、符号或条件差异，须回到英文原文核对。

Language / 词句

provided that

意为“只要”或“在……条件下”。阅读定理时，应把这类表达后面的内容识别为假设。

Idea / 难点

边注适合交代“这一步为什么成立”或提示先修知识。需要多行推导的证明补充放在正文，避免边栏过密。

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Preface to the second edition

Reading source

Lawrence C. Evans: original preface

English printed pp. xi–xii; PDF pages 12–13. Chinese reference: PDF page 9.

原著序言全文；中文边注为整理者补充，英文印次信息按所提供的英文 PDF 保留。

Let me thank everyone who over the past decade has provided me with suggestions and corrections for improving the first edition of this book. I am extraordinarily grateful. Although I have not always followed these many pieces of advice and criticism, I have thought carefully about them all. So many people have helped me out that it is unfortunately no longer feasible to list all their names. I have also received great help from everyone at the AMS, especially Sergei Gelfand, Stephen Moye and Arlene O'Sean. The NSF has generously supported my research during the writing of both the original edition of the book and this revision. I will continue to maintain lists of errors on my homepage, accessible through the `math.berkeley.edu` website.

When you write a big book on a big subject, the temptation is to include everything. A critic famously once imagined Tolstoy during the writing of *War and Peace*: "The book is long, but even if it were twice as long, if it were three times as long, there would always be scenes that have been omitted, and these Tolstoy, waking up in the middle of the night, must have regretted. There must have been a night when it occurred to him that he had not included a yacht race…" (G. Moore, *Avowals*).

This image notwithstanding, I have tried to pack into this second edition as many fascinating new topics in partial differential equations (PDE) as I could manage, most notably in the new Chapter 12 on nonlinear wave equations. There are new sections on Noether's Theorem and on local minimizers in the calculus of variations, on the Radon transform, on Turing instabilities for reaction-diffusion systems, etc. I have rewritten and expanded the previous discussions on blow-up of solutions, on group and phase velocities, and on several further subjects. I have also updated and greatly increased citations to books in the bibliography and have moved references to research articles to within the text. There are countless further minor modifications in notation and wording. Most importantly, I

Language / 词句

Although …, I have …

Although 引导让步："虽然没有全部采纳，但每条建议都认真考虑过。" AMS 为美国数学学会，NSF 为美国国家科学基金会。

中文提示

这里借托尔斯泰和"游艇赛"自嘲：再厚的书也无法囊括一切。第 10 章习题中的游艇赛会呼应这段话。

Language / 词句

notwithstanding

意为"尽管如此"。blow-up：解发生爆破；group / phase velocity：群速度 / 相速度。

have added about 80 new exercises, most quite interesting and some rather elaborate. There are now over 200 in total.

And there is a yacht race among the problems for Chapter 10.

LCE
January, 2010
Berkeley

This third printing of the second edition corrects all known mistakes, as of February, 2022.

中文提示

印次差异：英文此处为 2022 年；中译本对应写 2015 年，并附译者注说明参考了 2020 年勘误。此处按英文 PDF 保留，不能混作同一印次。

Preface to the first edition

Reading source

Lawrence C. Evans: original preface

English printed pp. xiii–xv; PDF pages 14–16. Chinese reference: PDF pages 11–12.

原著序言全文；中文边注为整理者补充。保留作者的写作原则、致谢和署名。

I present in this book a wide-ranging survey of many important topics in the theory of partial differential equations (PDE), with particular emphasis on various modern approaches. I have made a huge number of editorial decisions about what to keep and what to toss out, and can only claim that this selection seems to me about right. I of course include the usual formulas for solutions of the usual linear PDE, but also devote large amounts of exposition to energy methods within Sobolev spaces, to the calculus of variations, to conservation laws, etc.

Language / 词句

survey; exposition

survey 在此指综述；exposition 指系统讲解。全书既讲显式解公式，也讲能量方法、变分法与守恒律。

My general working principles in the writing have been these:

a. PDE theory is (mostly) not restricted to two independent variables. Many texts describe PDE as if functions of the two variables (x, y) or (x, t) were all that matter. This emphasis seems to me misleading, as modern discoveries concerning many types of equations, both linear and nonlinear, have allowed for the rigorous treatment of these in any number of dimensions. I also find it unsatisfactory to "classify" partial differential equations: this is possible in two variables, but creates the false impression that there is some kind of general and useful classification scheme available in general.

Idea / 难点

作者提醒：偏微分方程不限于两个自变量，也研究高维问题。他质疑的是把二维分类当作普适框架，并非否定具体方程的分类。

b. Many interesting equations are nonlinear. My view is that overall we know too much about linear PDE and too little about nonlinear PDE. I have accordingly introduced nonlinear concepts early in the text and have tried hard to emphasize everywhere nonlinear analogues of the linear theory.

Language / 词句

accordingly; analogues

accordingly：因此、相应地。nonlinear analogues 指线性理论在非线性问题中的相应思想或结果；能否推广仍须逐项证明。

c. Understanding generalized solutions is fundamental. Many of the partial differential equations we study, especially nonlinear first-order equations, do not in general possess smooth solutions. It is therefore essential to devise some kind of proper notion of generalized or weak solution. This is an important but subtle undertaking, and much of the hardest material in this book concerns the uniqueness of appropriately defined weak solutions.

d. PDE theory is not a branch of functional analysis. Whereas certain classes of equations can profitably be viewed as generating abstract operators between Banach spaces, the insistence on an overly abstract viewpoint, and consequent ignoring of deep calculus and measure theoretic estimates, is ultimately limiting.

e. Notation is a nightmare. I have really tried to introduce consistent notation, which works for all the important classes of equations studied. This attempt is sometimes at variance with notational conventions within a given subarea.

f. Good theory is (almost) as useful as exact formulas. I incorporate this principle into the overall organization of the text, which is subdivided into three parts, roughly mimicking the historical development of PDE theory itself. Part I concerns the search for explicit formulas for solutions, and Part II the abandoning of this quest in favor of general theory asserting the existence and other properties of solutions for linear equations. Part III is the mostly modern endeavor of fashioning general theory for important classes of nonlinear PDE.

Let me also explicitly comment here that I intend the development within each section to be rigorous and complete (exceptions being the frankly heuristic treatment of asymptotics in §4.5 and an occasional reference to a research paper). This means that even locally within each chapter the topics do not necessarily progress logically from "easy" to "hard" concepts. There are many difficult proofs and computations early on, but as compensation many easier ideas later. The student should certainly omit on first reading some of the more arcane proofs.

I wish next to emphasize that this is a textbook, and not a reference book. I have tried everywhere to present the essential ideas in the clearest possible settings, and therefore have almost never established sharp versions of any of the theorems. Research articles and advanced monographs, many of them listed in the Bibliography, provide such precision and generality. My goal has rather been to explain, as best I can, the many fundamental ideas of the subject within fairly simple contexts.

I have greatly profited from the comments and thoughtful suggestions of many of my colleagues, friends and students, in particular: S. Antman,

Idea / 难点

"弱化解的定义"与"保持唯一性"必须一起考虑。弱解、熵解、黏性解有各自的定义和适用情形，不能简单混用。

中文提示

作者强调不能只依靠抽象算子框架；具体的微积分计算、测度论和估计同样关键。这并不排斥使用泛函分析。

Language / 词句

at variance with

意为"与……不一致"。本书为保持全书统一，部分符号可能不同于某个专门领域的惯例；阅读时先核对附录 A。

中文提示

三部分的主线：先求显式公式，再研究线性方程的一般理论，最后研究若干重要非线性方程。存在性、唯一性等结论本身也是"求解"的成果。

Language / 词句

heuristic / sharp

heuristic：启发式的；arcane：深奥的；sharp：条件、常数等更精确或接近最优。作者优先解释思想；初读可暂缓部分难证明，不能把启发式推导当作完整证明。

J. Bang, X. Chen, A. Chorin, M. Christ, J. Cima, P. Colella, J. Cooper, M. Crandall, B. Driver, M. Feldman, M. Fitzpatrick, R. Gariepy, J. Goldstein, D. Gomes, O. Hald, W. Han, W. Hrusa, T. Ilmanen, I. Ishii, I. Israel, R. Jerrard, C. Jones, B. Kawohl, S. Koike, J. Lewis, T.-P. Liu, H. Lopes, J. McLaughlin, K. Miller, J. Morford, J. Neu, M. Portilheiro, J. Ralston, F. Rezakhanlou, W. Schlag, D. Serre, P. Souganidis, J. Strain, W. Strauss, M. Struwe, R. Temam, B. Tvedt, J.-L. Vazquez, M. Weinstein, P. Wolfe, and Y. Zheng.

I especially thank Tai-Ping Liu for many years ago writing out for me the first draft of what is now Chapter 11.

I am extremely grateful for the suggestions and lists of mistakes from earlier drafts of this book sent to me by many readers, and I encourage others to send me their comments, at `evans@math.berkeley.edu`. I have come to realize that I must be more than slightly mad to try to write a book of this length and complexity, but I am not yet crazy enough to think that I have made no mistakes. I will therefore maintain a listing of errors which come to light, and will make this accessible through the `math.berkeley.edu` homepage.

Faye Yeager at UC Berkeley has done a really magnificent job typing and updating these notes, and Jaya Nagendra heroically typed an earlier version at the University of Maryland. My deepest thanks to both.

I have been supported by the NSF during much of the writing, most recently under grant DMS-9424342.

LCE
August, 1997
Berkeley

Chapter 1

Introduction

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 1; printed p. 1; PDF page 18.

Chinese reference: 中译本对应章：引言，书内第 1 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

1.1 Partial Differential Equations

Outline only. Text, annotations, and proof details are pending.

1.2 Examples

Outline only. Text, annotations, and proof details are pending.

1.3 Strategies for Studying PDE

Outline only. Text, annotations, and proof details are pending.

1.4 Overview

Outline only. Text, annotations, and proof details are pending.

1.5 Problems

Outline only. Text, annotations, and proof details are pending.

1.6 References

Outline only. Text, annotations, and proof details are pending.

Part I

Representation Formulas for Solutions

Chapter 2

Four Important Linear Partial Differential Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 2; printed p. 17; PDF page 34.

Chinese reference: 中译本对应章：四个重要的线性偏微分方程，书内第 13 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

2.1 Transport Equation

Outline only. Text, annotations, and proof details are pending.

2.2 Laplace's Equation

Outline only. Text, annotations, and proof details are pending.

2.3 Heat Equation

Outline only. Text, annotations, and proof details are pending.

2.4 Wave Equation

Outline only. Text, annotations, and proof details are pending.

2.5 Problems

Outline only. Text, annotations, and proof details are pending.

2.6 References

Outline only. Text, annotations, and proof details are pending.

Chapter 3

Nonlinear First-Order PDE

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 3; printed p. 89; PDF page 106.

Chinese reference: 中译本对应章：非线性一阶偏微分方程，书内第 61 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

3.1 Complete Integrals, Envelopes

Outline only. Text, annotations, and proof details are pending.

3.2 Characteristics

Outline only. Text, annotations, and proof details are pending.

3.3 Introduction to Hamilton–Jacobi Equations

Outline only. Text, annotations, and proof details are pending.

3.4 Introduction to Conservation Laws

Outline only. Text, annotations, and proof details are pending.

3.5 Problems

Outline only. Text, annotations, and proof details are pending.

3.6 References

Outline only. Text, annotations, and proof details are pending.

Chapter 4

Other Ways to Represent Solutions

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 4; printed p. 159; PDF page 176.

Chinese reference: 中译本对应章：解的其他表示方法，书内第 111 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

4.1 Separation of Variables

Outline only. Text, annotations, and proof details are pending.

4.2 Similarity Solutions

Outline only. Text, annotations, and proof details are pending.

4.3 Transform Methods

Outline only. Text, annotations, and proof details are pending.

4.4 Converting Nonlinear into Linear PDE

Outline only. Text, annotations, and proof details are pending.

4.5 Asymptotics

Outline only. Text, annotations, and proof details are pending.

4.6 Power Series

Outline only. Text, annotations, and proof details are pending.

4.7 Problems

Outline only. Text, annotations, and proof details are pending.

4.8 References

Outline only. Text, annotations, and proof details are pending.

Part II

Theory for Linear Partial Differential Equations

Chapter 5

Sobolev Spaces

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 5; printed p. 239; PDF page 256.

Chinese reference: 中译本对应章：Sobolev 空间，书内第 169 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

5.1 Hölder Spaces

Outline only. Text, annotations, and proof details are pending.

5.2 Sobolev Spaces

Outline only. Text, annotations, and proof details are pending.

5.3 Approximation

Outline only. Text, annotations, and proof details are pending.

5.4 Extensions

Outline only. Text, annotations, and proof details are pending.

5.5 Traces

Outline only. Text, annotations, and proof details are pending.

5.6 Sobolev Inequalities

Outline only. Text, annotations, and proof details are pending.

5.7 Compactness

Outline only. Text, annotations, and proof details are pending.

5.8 Additional Topics

Outline only. Text, annotations, and proof details are pending.

5.9 Other Spaces of Functions

Outline only. Text, annotations, and proof details are pending.

5.10 Problems

Outline only. Text, annotations, and proof details are pending.

5.11 References

Outline only. Text, annotations, and proof details are pending.

Chapter 6

Second-Order Elliptic Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 6; printed p. 293; PDF page 310.

Chinese reference: 中译本对应章：二阶椭圆型方程，书内第 208 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

6.1 Definitions

Outline only. Text, annotations, and proof details are pending.

6.2 Existence of Weak Solutions

Outline only. Text, annotations, and proof details are pending.

6.3 Regularity

Outline only. Text, annotations, and proof details are pending.

6.4 Maximum Principles

Outline only. Text, annotations, and proof details are pending.

6.5 Eigenvalues and Eigenfunctions

Outline only. Text, annotations, and proof details are pending.

6.6 Problems

Outline only. Text, annotations, and proof details are pending.

6.7 References

Outline only. Text, annotations, and proof details are pending.

Chapter 7

Linear Evolution Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 7; printed p. 349; PDF page 366.

Chinese reference: 中译本对应章：线性发展方程，书内第 248 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

7.1 Second-Order Parabolic Equations

Outline only. Text, annotations, and proof details are pending.

7.2 Second-Order Hyperbolic Equations

Outline only. Text, annotations, and proof details are pending.

7.3 Systems of Hyperbolic First-Order Equations

Outline only. Text, annotations, and proof details are pending.

7.4 Semigroup Theory

Outline only. Text, annotations, and proof details are pending.

7.5 Problems

Outline only. Text, annotations, and proof details are pending.

7.6 References

Outline only. Text, annotations, and proof details are pending.

Part III

Theory for Nonlinear Partial Differential Equations

Chapter 8

The Calculus of Variations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 8; printed p. 427; PDF page 444.

Chinese reference: 中译本对应章：变分法，书内第 303 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

8.1 Introduction

Outline only. Text, annotations, and proof details are pending.

8.2 Existence of Minimizers

Outline only. Text, annotations, and proof details are pending.

8.3 Regularity

Outline only. Text, annotations, and proof details are pending.

8.4 Constraints

Outline only. Text, annotations, and proof details are pending.

8.5 Critical Points

Outline only. Text, annotations, and proof details are pending.

8.6 Invariance, Noether's Theorem

Outline only. Text, annotations, and proof details are pending.

8.7 Problems

Outline only. Text, annotations, and proof details are pending.

8.8 References

Outline only. Text, annotations, and proof details are pending.

Chapter 9

Nonvariational Techniques

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 9; printed p. 495; PDF page 512.

Chinese reference: 中译本对应章：非变分技术，书内第 350 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

9.1 Monotonicity Methods

Outline only. Text, annotations, and proof details are pending.

9.2 Fixed Point Methods

Outline only. Text, annotations, and proof details are pending.

9.3 Method of Subsolutions and Supersolutions

Outline only. Text, annotations, and proof details are pending.

9.4 Nonexistence of Solutions

Outline only. Text, annotations, and proof details are pending.

9.5 Geometric Properties of Solutions

Outline only. Text, annotations, and proof details are pending.

9.6 Gradient Flows

Outline only. Text, annotations, and proof details are pending.

9.7 Problems

Outline only. Text, annotations, and proof details are pending.

9.8 References

Outline only. Text, annotations, and proof details are pending.

Chapter 10

Hamilton–Jacobi Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 10; printed p. 543; PDF page 560.

Chinese reference: 中译本对应章：Hamilton–Jacobi 方程，书内第 384 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

10.1 Introduction, Viscosity Solutions

Outline only. Text, annotations, and proof details are pending.

10.2 Uniqueness

Outline only. Text, annotations, and proof details are pending.

10.3 Control Theory, Dynamic Programming

Outline only. Text, annotations, and proof details are pending.

10.4 Problems

Outline only. Text, annotations, and proof details are pending.

10.5 References

Outline only. Text, annotations, and proof details are pending.

Chapter 11

Systems of Conservation Laws

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 11; printed p. 571; PDF page 588.

Chinese reference: 中译本对应章：守恒律组，书内第 404 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

11.1 Introduction

Outline only. Text, annotations, and proof details are pending.

11.2 Riemann's Problem

Outline only. Text, annotations, and proof details are pending.

11.3 Systems of Two Conservation Laws

Outline only. Text, annotations, and proof details are pending.

11.4 Entropy Criteria

Outline only. Text, annotations, and proof details are pending.

11.5 Problems

Outline only. Text, annotations, and proof details are pending.

11.6 References

Outline only. Text, annotations, and proof details are pending.

Chapter 12

Nonlinear Wave Equations

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Chapter 12; printed p. 617; PDF page 634.

Chinese reference: 中译本对应章：非线性波动方程，书内第 437 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

12.1 Introduction

Outline only. Text, annotations, and proof details are pending.

12.2 Existence of Solutions

Outline only. Text, annotations, and proof details are pending.

12.3 Semilinear Wave Equations

Outline only. Text, annotations, and proof details are pending.

12.4 Critical Power Nonlinearity

Outline only. Text, annotations, and proof details are pending.

12.5 Nonexistence of Solutions

Outline only. Text, annotations, and proof details are pending.

12.6 Problems

Outline only. Text, annotations, and proof details are pending.

12.7 References

Outline only. Text, annotations, and proof details are pending.

Appendices

Appendix

A

Notation

> Reading source
> **Evans, *Partial Differential Equations*, second edition**
> English source: Appendix A; printed p. 653; PDF page 670.
> Chinese reference: 中译本对应章：记号，书内第 463 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

A.1 Notation for matrices

Outline only. Text, annotations, and proof details are pending.

A.2 Geometric notation

Outline only. Text, annotations, and proof details are pending.

A.3 Notation for functions

Outline only. Text, annotations, and proof details are pending.

A.4 Vector-valued functions

Outline only. Text, annotations, and proof details are pending.

A.5 Notation for estimates

Outline only. Text, annotations, and proof details are pending.

A.6 Some comments about notation

Outline only. Text, annotations, and proof details are pending.

Appendix

B

Inequalities

> Reading source
>
> **Evans, *Partial Differential Equations*, second edition**
>
> English source: Appendix B; printed p. 661; PDF page 678.
>
> Chinese reference: 中译本对应章：不等式，书内第 471 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

B.1 Convex functions

Outline only. Text, annotations, and proof details are pending.

B.2 Useful inequalities

Outline only. Text, annotations, and proof details are pending.

Appendix

C

Calculus

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix C; printed p. 665; PDF page 682.

Chinese reference: 中译本对应章：微积分，书内第 475 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

C.1 Boundaries

Outline only. Text, annotations, and proof details are pending.

C.2 Gauss–Green Theorem

Outline only. Text, annotations, and proof details are pending.

C.3 Polar coordinates, coarea formula

Outline only. Text, annotations, and proof details are pending.

C.4 Moving regions

Outline only. Text, annotations, and proof details are pending.

C.5 Convolution and smoothing

Outline only. Text, annotations, and proof details are pending.

C.6 Inverse Function Theorem

Outline only. Text, annotations, and proof details are pending.

C.7 Implicit Function Theorem

Outline only. Text, annotations, and proof details are pending.

C.8 Uniform convergence

Outline only. Text, annotations, and proof details are pending.

Appendix

D

Functional Analysis

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix D; printed p. 674; PDF page 691.

Chinese reference: 中译本对应章：泛函分析，书内第 482 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

D.1 Banach spaces

Outline only. Text, annotations, and proof details are pending.

D.2 Hilbert spaces

Outline only. Text, annotations, and proof details are pending.

D.3 Bounded linear operators

Outline only. Text, annotations, and proof details are pending.

D.4 Weak convergence

Outline only. Text, annotations, and proof details are pending.

D.5 Compact operators, Fredholm theory

Outline only. Text, annotations, and proof details are pending.

D.6 Symmetric operators

Outline only. Text, annotations, and proof details are pending.

Appendix

E

Measure Theory

Reading source

Evans, *Partial Differential Equations*, second edition

English source: Appendix E; printed p. 684; PDF page 701.

Chinese reference: 中译本对应章：测度论，书内第 490 页起。

Status: outline only / 目录骨架。 English exposition, Chinese margin notes, and expanded proofs will be added section by section.

E.1 Lebesgue measure

Outline only. Text, annotations, and proof details are pending.

E.2 Measurable functions and integration

Outline only. Text, annotations, and proof details are pending.

E.3 Convergence theorems for integrals

Outline only. Text, annotations, and proof details are pending.

E.4 Differentiation

Outline only. Text, annotations, and proof details are pending.

E.5 Banach space-valued functions

Outline only. Text, annotations, and proof details are pending.

Bibliography

The source bibliography starts on English printed p. 689 (PDF page 706). Bibliographic entries and any additional references remain to be entered.

Outline only. Text, annotations, and proof details are pending.

Index

The source index starts on English printed p. 703 (PDF page 720). An index for these notes will be prepared as the text is written; page numbers must refer to this edition.

Outline only. Text, annotations, and proof details are pending.

Bilingual Glossary
英汉术语表

This added study aid will collect the English terms used in the notes, their Chinese equivalents, and links to the relevant definitions. Entries remain to be prepared.

Outline only. Text, annotations, and proof details are pending.